SIYAMTHANDA AMAHLE BUTHELEZI

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PROFILE

Data & BI Analytics Engineer with 2 years of experience specializing in Salesforce engineering, automated ETL pipeline development, and commercial analytics. Proficient in SQL, Python (Pandas), and Power BI (Star Schema, DAX), architecting end-to-end data workflows that resolve systemic data quality failures and ensure rigorous data integrity. Proven ability to audit data infrastructure to recover obscured revenue, and develop predictive churn and financial forecast models that drive strategic decision-making.

KEY SKILLS

- **Systems Operations & CRM:** Salesforce Administration, Native Cloud Architecture, SOQL, Apex, Flows, Synatic (Integration & ETL Monitoring), Power Automate, Root Cause Analysis, Bulk Data Operations.
- **Data Engineering & Scripting:** Python (Pandas, XlsxWriter, NumPy, Scikit-Learn), SQL, Advanced Excel, Modular ETL Pipeline Development, Data Integrity & Edge Case Handling, Automated Data Masking & Compliance (POPIA), Git.
- **Data Science & ML:** Predictive Modeling (Random Forest, Logistic Regression), Feature Engineering, Time Series Forecasting (Holt-Winters), Unsupervised Learning (K-Means, PCA).
- **Visualization & BI:** Power BI (Star Schema Data Modeling, DAX, Secure Publishing), Tableau, Data Storytelling, Matplotlib, Seaborn.
- **Soft Skills:** Stakeholder Management, Technical Runbook Documentation, Agile Methodologies.

EDUCATION

Bachelor of Commerce in Management Information Systems, University of Zululand,
February 2022 — November 2024
Majors: Information Systems and Computer Science

EXPERIENCE

Data & BI Analytics Engineer at Medecins Sans Frontieres (Southern Africa),
January 2025 — Present

- **Predictive Modeling & Machine Learning:** Developed a predictive churn model using Python (Scikit-Learn) to flag at-risk donors. Engineered 15+ behavioural features (giving frequency, lost opportunities) to achieve 97% recall, generating 0-100 risk scores for 16K+ active profiles to guide proactive retention strategies.
- **Root Cause Analysis & BI:** Analyzed 96K+ cancellation events over 3 years, deduplicating to isolate 17K+ unique donor losses. Built and securely published interactive Power BI dashboards uncovering critical anomalies, and utilizing Power Query to map donor lifecycles, identifying a high-risk Onboarding Cliff.
- **Reactivation ROI & Failure Analytics:** Engineered a modular Python ETL pipeline (Pandas) to merge 4 relational Salesforce exports, tracking 1st/2nd debit success rates and lifetime revenue for a 1,4K+ donor reactivation cohort. Audited and resolved systemic silent merge failures (data type coercion, whitespace stripping, inconsistent ID hyphens) suppressing R400K+ in invisible revenue and misreporting campaign success rates (correcting 1st debit rate from 5% to 53%). Architected an interactive Power BI dashboard (Star Schema, DAX) visualizing KPIs, lifecycle success rates, and failure root causes, enabling targeted follow-up strategies.
- **Enterprise BI & Workflow Automation:** Migrated a localized Python data-cleaning pipeline to a cloud-integrated Excel/SharePoint workflow, eliminating the need for non-technical end-users to execute code and reducing monthly reporting time to a single click.
- **Financial Forecasting:** Tasked by Director to develop a 10-year financial forecast and strategic model for the fundraising department, supporting long-term budgeting, scenario analysis, and operational planning.

- **Automated ETL & Data Privacy:** Engineered robust Python data pipelines (Pandas, XlsxWriter) to extract, clean, and transform raw CRM exports. Implemented automated PII masking algorithms for sensitive data (Credit Cards, IDs) to maintain strict POPIA compliance prior to system ingestion.
- **Cloud-Native ETL & Process Automation:** Evolved operational workflows from manual Excel processes to Python automations, and ultimately to a secure, zero-dependency Salesforce architecture. Architected a serverless pipeline (Scheduled Flow & Apex) that autonomously queries relational data, applies dynamic in-memory PII masking, executes complex business logic, and auto-distributes secure CSVs to stakeholders eliminating local security vulnerabilities.
- **Process Optimization:** Engineered an end-to-end request management system using Power Automate and SharePoint, featuring role-based routing and automated notifications, reducing administrative overhead.
- **Runbooks & Operational Documentation:** Authored comprehensive technical Runbooks and Standard Operating Procedures (SOPs) for complex data pipelines and ML workflows, ensuring seamless operational continuity and minimal downtime during handovers.
- **System Monitoring:** Proactively monitored batch transaction pipelines (Synatic/LinkServ), identifying and resolving data interruptions to ensure continuous, automated system processing.

PROJECTS AND VIRTUAL EXPERIENCE

Data Analyst at Quantium/Forage Virtual Programme
March 2026

- Completed a job simulation focused on Data Analytics and Commercial Insights for the data science team.
- Developed expertise in data preparation and customer analytics, utilizing transaction datasets to extract valuable insights and deliver data-driven commercial recommendations.
- Extended analytical capabilities to identify benchmark stores for conducting uplift testing on trial store layouts, enabling evidence-based decision-making.
- Leveraged acquired data analytics and insights from previous tasks to create comprehensive reports for the Category Manager, facilitating informed strategic decisions and enhancing commercial applications.

Data Analyst at Deloitte Australia/Forage Virtual Programme
March 2026

- Completed a Deloitte job simulation involving data analysis and forensic technology.
- Created a data dashboard using Tableau and used Excel to classify data and draw business conclusions.

Introductory Data Scientist at Commonwealth Bank/Forage Virtual Programme
August 2025

- Completed a job simulation involving Data Management skills for Commonwealth's Data Science Team.
- Demonstrated proficiency in creating data engineering pipelines to aggregate and extract valuable insights from datasets, optimizing data-driven decision-making.
- Acquired skills in anonymizing personal data within datasets, ensuring compliance with data privacy regulations.
- Proposed effective data analysis approaches, particularly related to social media, and demonstrated the ability to design well-structured databases for efficient information management.

Student Fingerprint Verification System at University of Zululand
Capstone Project | June 2024 – November 2024

- Spearheaded requirements documentation (SRS) and system design documentation (SDD) for a biometric student attendance system, translating stakeholder needs into technical specifications.
- Researched and implemented Support Vector Machine (SVM) fingerprint recognition using Python libraries (scikit-learn, OpenCV), achieving exceptional performance metrics:
  - **Precision: 93.82%** - **Accuracy: 89%** - **Recall: 83.5%**
- Navigated project constraints when institutional budget limitations prevented acquisition of the fingerprint scanner SDK, requiring rapid adaptation of demonstration approach.
- Engineered a simulated environment to showcase system functionality to stakeholders, including university administrators and faculty, demonstrating core features and value proposition.

- Presented system architecture, model performance metrics, and implementation roadmap to a live audience, receiving positive feedback on analytical approach and problem-solving adaptability.
- Documented detailed requirements for full implementation, including SDK specifications and integration requirements, to enable future deployment.

PERSONAL DATA SCIENCE PROJECTS

South Africa Inflation Analysis Trilogy (Python, Pandas, Statsmodels)

- Forecasting: Built a Holt-Winters Exponential Smoothing model to forecast 2025-2026 inflation rates, achieving a Mean Absolute Error (MAE) of 0.43% against economist projections.
- Volatility Analysis: Analysed 50 years of economic data to identify structural breaks and "eras of chaos," calculating rolling volatility to correlate historical events (e.g., 1994 Elections, 2008 Crisis) with market instability.
- Purchasing Power Simulation: Engineered a simulation model to compare nominal wage growth against cumulative inflation, visualizing the "silent erosion" of real wages and the long-term impact of economic trends on consumer purchasing power.

Global Homicide Trends Analysis [1970–2030] (Python, Scikit-Learn, Statsmodels)

- Unsupervised Learning (Clustering): Applied K-Means Clustering to 13 nations to segment them into distinct "Crime Trajectories" (Safe Havens vs. Critical Zones), identifying South Africa and Colombia as distinct outliers.
- Time Series Forecasting: Utilized Holt-Winters Exponential Smoothing to project homicide rates for South Africa and Colombia into 2030, predicting stagnation and slight increase respectively.
- Divergence Analysis: Visualized the "Global Safety Gap" over 50 years, revealing persistent inequality between the safest (Japan/Singapore) and most dangerous nations.

Walsoft Institute: Student Success Optimization (Python, Pandas, Scikit-Learn)

- Strategic Analysis: Conducted a full-cycle consulting analysis to optimize student performance and resource allocation for a Business Intelligence training program, analyzing demographic, academic, and behavioral data.
- Data Wrangling: Engineered a robust ETL pipeline to clean a messy raw dataset, resolving inconsistencies in categorical data (Country, Residence, Gender) and handling missing values via imputation.
- Predictive Modelling: Built a Random Forest Regressor to identify performance drivers, uncovering that Study Hours drive 68% of Python success (Effort-based) while Entry Exams drive 60% of DB success (Aptitude-based).
- Business Impact & ROI: Developed a segmentation strategy identifying at-risk groups (e.g., specific residence types) and recommended a "Gamified Study Hour" intervention as the highest-ROI strategy to improve pass rates.
- Deliverables: Automated the generation of executive reports, strategic recommendations, and visualization dashboards (Correlation Heatmaps, Feature Importance plots) for stakeholder review.